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Z80 HARDWARE ORGANIZATION

INTRODUCTION

In order to program at an elementary level, it is not necessary to understand in detail the internal structure of the processor that one is using. However, in order to do efficient programming, such an understanding is required. The purpose of this chapter is to present the basic hardware concepts necessary for understanding the operation of the Z80 system. The complete microcomputer system includes not only the microprocessor unit (here the Z80), but also other components. This chapter presents the Z80 proper, while the other devices (mainly input/output) will be presented in a separate chapter (Chapter 7).

We will review here the basic architecture of the microcomputer system, then study more closely the internal organization of the Z80. We will examine, in particular, the various registers. We will then study the program execution and sequencing mechanism. From a hardware standpoint, this chapter is only a simplified presentation. The reader interested in gaining detailed understanding is referred to our book ref. C201 ("Microprocessors," by the same author).

The Z80 was designed as a replacement for the Intel 8080, and to offer additional capabilities. A number of references will be made in this chapter to the 8080 design.

SYSTEM ARCHITECTURE

The architecture of the microcomputer system appears in Figure 2.1. The microprocessor unit (MPU), which will be a Z80 here, appears on the left of the illustration. It implements the functions of a *central-processing unit* (CPU) within one chip: it includes an *arithmetic-logical unit* (ALU), plus its internal registers, and a *control unit* (CU), in